

IoT Challenge 3 (LoRaWAN Exercise)

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1 EQ1 – Largest spreading factor for PSR $\geq 75\%$

Network parameters

- Carrier frequency: 868 MHz
- Bandwidth: BW = 125 kHz
- Number of nodes: $N = 40$
- Traffic rate per node: $\lambda = 2$ packets/min = $1/30$ packets/s
- Payload size: PL = 20 bytes
- Coding rate 4/5 (CR parameter = 1)
- Explicit header, CRC enabled, 8-symbol preamble

Step 1 – Time on air

Using the standard LoRa airtime formula (see TTN Airtime Calculator):

```
T_symbol = 2^SF / BW

T_preamble = (n_preamble + 4.25) * T_symbol [n_preamble = 8]

n_payload = 8 + max( ceil((8*PL - 4*SF + 28 + 16*CRC - 20*IH)
                        / (4*(SF - 2*DE))) * (CR+4) , 0 )
                    where DE = 1 if SF >= 11, else 0 (Low Data-Rate Optimisation)

T_air = T_preamble + n_payload * T_symbol
```

Substituting PL= 20, CRC= 1, IH= 0, CR= 1, $n_{\text{preamble}} = 8$:

SF	T_{sym} (ms)	DE	n_{pay} (sym)	T_{air} (ms)
7	1.024	0	43	56.58
8	2.048	0	38	102.91
9	4.096	0	33	185.34
10	8.192	0	33	370.69
11	16.384	1	33	741.38
12	32.768	1	28	1318.91

Step 2 – Packet success rate (pure ALOHA model)

Under the simplified uncoordinated model, transmissions follow a Poisson process. Using the pure ALOHA approximation, a packet succeeds if no other transmission overlaps it within a vulnerability window of $2T_{\text{air}}$:

$$\begin{aligned} G &= N * \lambda * T_{\text{air}} \text{ (offered load, dimensionless)} \\ &= 40 * (2/60) * T_{\text{air}}(\text{s}) \\ \text{PSR} &= \exp(-2G) \end{aligned}$$

Threshold: $\text{PSR} \geq 0.75 \implies e^{-2G} \geq 0.75 \implies G \leq -\ln(0.75)/2 \approx 0.1438$. This corresponds to a maximum T_{air} of about **107.9 ms**.

SF	T_{air} (ms)	G	PSR (%)	PSR $\geq$ 75 %?
7	56.58	0.0754	86.00	✓ Yes
8	102.91	0.1372	76.00	✓ Yes
9	185.34	0.2471	61.00	× No
10	370.69	0.4943	37.21	× No
11	741.38	0.9885	13.85	× No
12	1318.91	1.7586	2.97	× No

Answer EQ1: The largest spreading factor that guarantees $\text{PSR} \geq 75\%$ is **SF 8** ($\text{PSR} \approx 76\%$). SF 9 and above produce too much airtime ($T_{\text{air}} > 107.9\text{ ms}$), pushing G above the threshold and reducing PSR below 75 %.

2 EQ2 – Why is measured performance lower and non-uniform?

Observation

After deployment, the measured PSR is significantly lower than the theoretical 75 % predicted for SF 8, and performance is *non-uniform*: some nodes achieve reasonable delivery rates while others experience very poor performance.

Root cause analysis

The pure ALOHA model assumes all nodes experience the *same channel quality* (same received SNR) and that every collision is fatal (no capture effect). In practice, two phenomena break this assumption:

- **Near-far problem / capture effect:** nodes physically closer to the gateway generate higher received signal power. When two packets collide, the gateway may still decode the stronger one (“capture”), favouring close nodes while starving distant ones.
- **Path loss and shadowing:** obstacles attenuate signals from far nodes below the demodulation threshold for SF 8, causing failures even without collisions.

Both effects match the symptom: overall PSR lower than expected *and* a wide spread across nodes.

Evaluation of the three options

#	Action	Effect on collisions	Effect on non-uniformity	Verdict
1	Decrease number of nodes	Reduces $G \Rightarrow$ fewer collisions	Does <i>not</i> equalise SNR; distant nodes still fail	Partial; does not fix non-uniformity
2	Move nodes closer to gateway	Relieves congestion indirectly	Equalises received power $\Rightarrow$ less near-far imbalance	Best: addresses both root causes
3	Increase spreading factor	Increases $T_{\text{air}} \Rightarrow$ <i>more</i> collisions	Helps link budget but worsens collision-limited throughput	Counter-productive

Why option 2 is correct: Moving nodes closer reduces path loss, especially at the cell edge, which (a) raises received SNR above the demodulation floor without necessarily increasing SF, (b) reduces the near-far power imbalance, and (c) allows ADR to use lower SF, shortening airtime. The outcome is higher and more uniform PSR.

Why option 3 is wrong: Higher SF improves sensitivity but, when collisions dominate, longer airtime increases G and collision probability across the entire network.

Answer EQ2: The most appropriate action is **Option 2 – move nodes closer to the gateway**.

Non-uniform performance points to near-far and path-loss effects. Bringing nodes closer lifts weak links, aligns received powers, and avoids the extra collision load caused by lengthening packets with higher SF (option 3).
